

Lecture 23

06/11/24 10:10 AM

$$P(X_{10}=j) = \sum_i P(X_{10}=j | X_0=i) \cdot P(X_0=i)$$

Classification of states

j is accessible from $i \Rightarrow p_{ij}^n > 0$ for some n
 (denoted by $i \rightarrow j$)
 $\Rightarrow P(X_n=j | X_0=i) > 0$ for some n

If $i \rightarrow j$ and $j \rightarrow i \Rightarrow i$ and j communicate
 (denoted by $i \leftrightarrow j$)

Irreducible chain $\Rightarrow i \leftrightarrow j \forall i, j \in S$

* Recurrent state

$F_{ii} = P(\text{ever returning to } i \text{ having started in } i) = 1$
 ↳ not easy to calculate

e.g. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

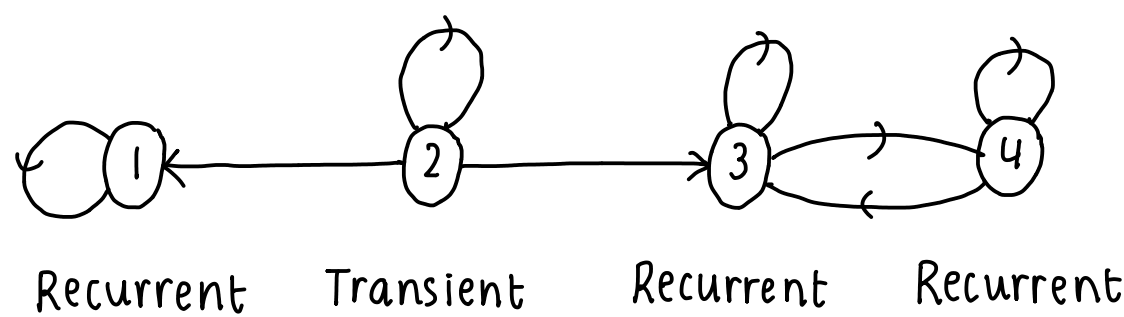
both states are recurrent

* Transient state

= not recurrent

$$F_{ii} < 1$$

* If $i \leftrightarrow j$ and i is recurrent $\Rightarrow j$ is recurrent



Limiting Probabilities

After some n , every row is the same.

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0.6 & 0.4 \\ 0.6 & 0.4 & 0 \end{bmatrix}, \quad P^5 = \begin{bmatrix} 0.06 & 0.30 & 0.64 \\ 0.18 & 0.38 & 0.44 \\ 0.38 & 0.44 & 0.18 \end{bmatrix}, \quad P^{30} = \begin{bmatrix} 0.23 & 0.385 & 0.385 \\ 0.23 & 0.385 & 0.385 \\ 0.23 & 0.385 & 0.385 \end{bmatrix}$$

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = [\lim_{n \rightarrow \infty} P^n]_{ij}$$

$\alpha_j = \lim_{n \rightarrow \infty} p_{ij}^{(n)}$ = probability of being in state j after a large time from starting in state i .

For an M state DTMC, $\alpha = (\alpha_1, \dots, \alpha_M)$ denotes the limiting distribution.

Stationary distribution

= solution to the equation $\pi = \pi P$

πP = pmf of X_1 having picked X_0 according to π .

$\therefore \pi = \pi P \Rightarrow$ if the initial distribution is π ,
 then the distribution of X_1 is also π .

Continuing this argument, the pmf of X_n for any n is π
 and there is no dependence on the starting state.

$$\pi = \pi P = (\pi P) P = \pi P^2 = \pi P^3 \dots$$

* The limiting distribution α need not exist for some Markov chains,
 but the stationary distribution exists.

* The limiting distribution if it exists,
 is the same as the stationary distribution.
 i.e. $\alpha_i = \pi_i \forall i$

!!! EXAMPLE: $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0.6 & 0.4 \\ 0.6 & 0.4 & 0 \end{bmatrix}$

$$\pi = \pi P \Rightarrow (\pi_1 \ \pi_2 \ \pi_3) = (\pi_1 \ \pi_2 \ \pi_3) \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0.6 & 0.4 \\ 0.6 & 0.4 & 0 \end{bmatrix}$$

$$\boxed{\sum \pi_i = 1} \quad \boxed{\pi_1 + \pi_2 + \pi_3 = 1} \quad \text{--- (1)}$$

$$\pi_1 = 0.6 \pi_3 \quad \text{--- (2)}$$

$$\begin{aligned} \pi_2 &= 0.6 \pi_2 + 0.4 \pi_3 \\ \Rightarrow 0.4 \pi_2 &= 0.4 \pi_3 \\ \Rightarrow \pi_2 &= \pi_3 \quad \text{--- (3)} \end{aligned}$$

$$\pi_3 = \pi_1 + 0.4 \pi_2 \quad \text{--- (4)}$$

$$\text{From (1), (2), (3)} \Rightarrow 0.6 \pi_3 + 2\pi_3 = 1$$

$$\pi_3 = \frac{1}{2.6} = \frac{10}{26}, \quad \pi_2 = \frac{10}{26}, \quad \pi_1 = \frac{6}{26}$$